```
clc; clear;
a = 5; b = 4; T1 = 80; max_term = 100
```

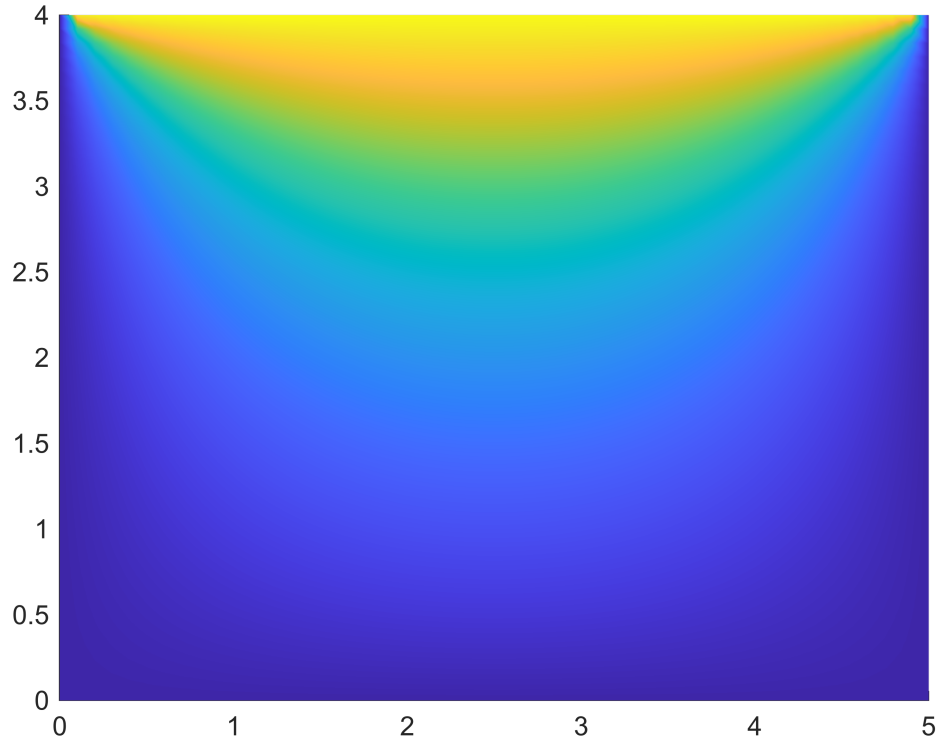
```
max_term =
100
```

```
syms x y n
T(x,y,n) = (4 * T1 / sym(pi)) * ...
    (sin((2 * n - 1)*(sym(pi) * x / a)) * sinh((2 * n - 1)*(sym(pi) * y / a))) / ...
    ((2 * n - 1) * sinh((2 * n - 1) * (sym(pi) * b / a)))
```

T(x, y, n) =

$$\frac{320 \sin\left(\frac{\pi\, x\, (2\, n - 1)}{5}\right) \sinh\left(\frac{\pi\, y\, (2\, n - 1)}{5}\right)}{\pi \sinh\left(\frac{4\, \pi\, (2\, n - 1)}{5}\right) (2\, n - 1)}$$

```
t = matlabFunction(T);

xspan = linspace(0, a, 50);
yspan = linspace(0, b, 50);

[X, Y] = meshgrid(xspan, yspan);

arr = zeros(length(yspan), length(xspan));

for n = 1 : max_term
    arr = arr + t(X, Y, n);
end

surf(X, Y, arr)
shading interp;
view(2)
```


4
3.5
3
2.5
2
1.5
1
0.5
0
0
1
2
3
4
5